

Problem Statement

Objective:

The goal of this analysis is to help customers identify the best credit card option based on their

spending habits and financial profile. We are provided with two datasets:

1. Bank Credit Card Data: This dataset includes information about various credit card options, including cashback percentages across different categories (grocery, travel, dining), annual fees, required income, and more.

2. Customer Data: This dataset includes information about individual customers, such as their annual income, monthly spending patterns, and preferences for cashback categories. Using these datasets, we want to develop a system that helps customers make informed decisions about which credit card suits their spending habits, budget, and financial goals.

Tables:

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```
bank_df = spark.read.format("delta").load("dbfs:/user/hive/warehouse/bank_data")
customer_df = spark.read.format("delta").load("dbfs:/user/hive/warehouse/customer_expense_dataset_1500")

bank_df.display()
customer_df.display()

bank_df.cache()
customer_df.cache()
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(2) Spark Jobs

Table

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	Bank_Name	Card_Name	Annual_Fee	Grocery_Cashback(%) [Zepto, BigBasket, etc.]
1	ICICI	platinum ship credit card	0	1.5
2	ICICI	Amazon pay icici card	0	3.0
3	ICICI	Coral Credit Card	500	5.0
4	IDFC	Millennia	0	1.0
5	IDFC	First Classic	0	0.0
6	IDFC	WOW ! Black	750	3.0
7	IDFC	First Select	0	1.0
8	IDFC	Ace	499	2.0
9	IDFC	Republic Day	0	3.0
10	PNB	Global Platinum	750	5.0
11	PNB	Global Gold	500	5.0
12	PNB	Global Classic	0	4.0

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	AC Name	123 Grocery	123 Travelling	123 Electricity	123 Internet	123 EMI	123 Dining	123 En						
1	Ishaan Sharma	2860	5390	4272	1544	6265	466							
2	Dev Desai	10322	1685	1269	843	2433	5311							
3	Aryan Chopra	10666	18942	2547	1999	2747	9167							
4	Aditya Kapoor	9734	1267	2028	1654	3556	3890							
5	Harsh Chopra	10792	10555	2561	741	11016	7513							
6	Aditya Kapoor	10226	14541	3652	1911	19457	1021							
7	Rohan Mehta	12173	995	3842	955	25851	1016							
8	Vivaan Mishra	14185	6331	879	992	39976	5276							
9	Vihaan Joshi	4027	2695	1995	1586	5258	5618							
10	Sai Pillai	7892	3561	2778	1517	38952	3099							
11	Sai Singh	14790	2454	4145	1304	30818	2731							
12	Reyansh Gupta	8910	12685	1370	1294	206	5134							

Use Cases

Here are 10 potential use cases that can be addressed using Python libraries like pandas

and numpy for data analysis:

1. Find the Best Credit Card for Grocery Shoppers

- Goal: Identify which credit card offers the highest cashback on grocery purchases (for example, from Zepto, BigBasket, etc.) based on the available percentage in the dataset.
- Input: Customer's monthly grocery spending.
- Output: The best credit card(s) based on grocery cashback rate.

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Python

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from pyspark.sql.functions import desc

best_grocery_cards = bank_df.orderBy(desc
("Grocery_Cashback")) \
    .select("Bank_Name",
            "Card_Name",
            "Grocery_Cashback")

best_grocery_cards.display(5)

```

▶ (1) Spark Jobs

	Bank_Name	Card_Name	.00 Grocery_Cashback
1	Bank of Baroda	IRCTC Credit Card	10.0
2	Bank of Baroda	HPCL Credit Card	10.0
3	Bank of Baroda	Prime Credit Card	10.0
4	Kotak Mahindra Bank	Kotak Cashback+ Credit Card	5.0
5	IDBI	IDBI Delight Credit Card	5.0
6	Axis Bank	Axis Bank Value+ Debit Card	5.0
7	HDFC Bank	Swiggy	5.0
8	SBI	Cashback SBI Card	5.0
9	PNB	Global Gold	5.0
10	PNB	Global Platinum	5.0
11	ICICI	Coral Credit Card	5.0
12	PNB	Millennial	4.0
13	PNB	Global Classic	4.0

2. Identify Cards with No Annual Fee

● Goal: Filter out credit cards that do not have an annual fee and show them to customers

who prefer no upfront costs.

● Input: Annual fee data.

● Output: List of credit cards with zero annual fee.

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10:54 AM (1s) 8 Python
no_fee_cards = bank_df.filter(col("Annual_Fee")
== 0) \
    .select("Bank_Name",
            "Card_Name")

no_fee_cards.display()
▶ (1) Spark Jobs

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	Table ▾	+
	Bank_Name	Card_Name
1	ICICI	platinum ship credit card
2	ICICI	Amazon pay icici card
3	IDFC	Millennia
4	IDFC	First Classic
5	IDFC	First Select
6	IDFC	Republic Day
7	PNB	Global Classic
8	PNB	Patanjali RuPay
9	Bank of Baroda	Prime Credit Card
10	Canara Bank	VISA Classic Credit Card
11	Canara Bank	MasterCard Standard

3. Match Customers with Cards Based on Income

- Goal: Identify credit cards that are available to a customer based on their income level.
- Input: Customer's annual income.
- Output: List of credit cards the customer qualifies for based on their income.

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11:29 AM (1s) 10 Python
from pyspark.sql.functions import broadcast, col

eligible_cards = customer_df.join(
    broadcast(bank_df),
    customer_df["Income"] >= bank_df
    ["Min_Income_Required(Yearly)"]
).select("Name", "Bank_Name", "Card_Name",
        "Income")

eligible_cards.display()

(2) Spark Jobs
eligible_cards.show()
DataFrame[Name: String, Bank_Name: String, Card_Name: String, Income: Double]

```

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	^A _C Name	^A _C Bank_Name	^A _C Card_Name	¹ ₃ Income	
1	Aditya Chopra	PNB	RuPay Select	56105	
2	Aarav Gupta	SBI	SBI Global International Debit Card	156063	
3	Ishaan Singh	SBI	SBI Platinum International Debit Card	80108	
4	Aryan Agarwal	Bank of Baroda	Prime Credit Card	132946	
5	Dev Reddy	Canara Bank	VISA Classic Credit Card	193707	
6	Arjun Mehta	Canara Bank	RuPay Debit Card	140627	
7	Kunal Nair	Canara Bank	VISA Debit Card	172038	
8	Reyansh Nair	Canara Bank	MasterCard Debit Card	157813	
9	Aryan Malhotra	Canara Bank	EMV Chip Debit Card	117235	
10	Kunal Mehta	Axis Bank	Axis Bank Delight Debit Card	41938	

4. Calculate Total Cashback for Each Customer

- Goal: Calculate the total cashback a customer would receive based on their spending across groceries, travel, and dining categories.
- Input: Customer's monthly spending in grocery, travel, and dining categories.
- Output: Total cashback value per credit card based on their spending.

```

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from pyspark.sql.functions import expr

cashback_df = customer_df.crossJoin(bank_df)

cashback_df = cashback_df.withColumn(
    "Total_Cashback",
    (col("Grocery") * col("Grocery_Cashback") /
     100) +
    (col("Travelling") * col("Travel_Cashback(%)
    ") / 100) +
    (col("Dining") * col("Dining_Cashback") /
     100)
)

cashback_df.select("Name", "Card_Name",
                  "Total_Cashback").display()

```

	Table ▾	+	
	⌵ ⌴ Name	⌵ ⌴ Card_Name	.00 Total_Cashback
1	Ishaan Sharma	platinum ship credit card	220.600000
2	Dev Desai	Amazon pay icici card	553.240000
3	Aryan Chopra	Coral Credit Card	4260.900000
4	Aditya Kapoor	Millennia	97.340000
5	Harsh Chopra	First Classic	0.000000
6	Aditya Kapoor	WOW ! Black	773.640000
7	Rohan Mehta	First Select	131.890000
8	Vivaan Mishra	Ace	452.530000
9	Vihaan Joshi	Republic Day	343.250000
10	Sai Pillai	Global Platinum	594.400000
11	Sai Singh	Global Gold	998.750000
12	Reyansh Gupta	Global Classic	1144.670000

5. Identify Cards with Best Dining Cashback

- Goal: Find the credit cards that offer the highest cashback on dining (e.g., through Swiggy, Zomato).
- Input: Customer's monthly dining spending.
- Output: List of cards with the highest dining cashback percentage.

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10:57 AM (<1s) 14
best_dining_cards = bank_df.orderBy(desc
("Dining_Cashback")) \
    .select("Bank_Name",
            "Card_Name",
            "Dining_Cashback")

best_dining_cards.display(5)
▶ (1) Spark Jobs

```

	Bank_Name	Card_Name	.00 Dining_Cashback
1	Axis Bank	Axis Bank Delight Debit Card	25.0
2	ICICI	Coral Credit Card	20.0
3	IDBI	Royale Signature card	15.0
4	ICICI	platinum ship credit card	15.0
5	IDBI	IDBI Delight Credit Card	15.0
6	HDFC Bank	Swiggy	10.0
7	HDFC Bank	Millennia	10.0
8	Axis Bank	Axis Bank Titanium Rewards Debit Card	10.0
9	HDFC Bank	Freedom	10.0
10	Axis Bank	Axis Bank Neo Credit Card	10.0
11	Axis Bank	Airtel Axis Bank Credit Card	10.0
12	HDFC Bank	Diners Club Privilege	5.0

6. Card Comparison: Annual Fee vs. Cashback

- Goal: Compare the annual fee of credit cards with the total cashback value a customer would get.
- Input: Customer's total annual spending in different categories.
- Output: List of cards with the best value when considering both the annual fee and cashback.

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10:58 AM (1s) 16
comparison_df = cashback_df.withColumn(
    "Net_Value",
    col("Total_Cashback") - col("Annual_Fee")
)

comparison_df.select("Name", "Card_Name",
    "Annual_Fee", "Total_Cashback", "Net_Value") \
    .orderBy(desc("Net_Value")).display
()
(2) Spark Jobs

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	Bank Name	Card Name	Annual_Fee	Total_Cashback	Net_Value
1	Atharva Rao	Coral Credit Card	500	5518.300000	5018.300000
2	Ayaan Gupta	Coral Credit Card	500	5451.700000	4951.700000
3	Sai Iyer	Coral Credit Card	500	5416.850000	4916.850000
4	Ishaan Gupta	Coral Credit Card	500	5360.250000	4860.250000
5	Reyansh Sharma	Coral Credit Card	500	5328.950000	4828.950000
6	Reyansh Bansal	Coral Credit Card	500	5293.250000	4793.250000
7	Ishaan Pillai	Coral Credit Card	500	5292.350000	4792.350000
8	Aditya Sharma	Coral Credit Card	500	5289.750000	4789.750000
9	Ishaan Gupta	Coral Credit Card	500	5272.000000	4772.000000
10	Rohan Verma	Coral Credit Card	500	5259.950000	4759.950000
11	Reyansh Mishra	Coral Credit Card	500	5258.050000	4758.050000
12	Kunal Joshi	Coral Credit Card	500	5227.850000	4727.850000
13	Kunal Gupta	Coral Credit Card	500	5226.600000	4726.600000
14	Kabir Rao	Coral Credit Card	500	5223.250000	4723.250000

7. Filter Cards Based on Monthly Spending Limit

- Goal: Identify cards that offer a monthly spending limit suitable for the customer's financial needs.
- Input: Customer's desired monthly credit limit.
- Output: List of credit cards with monthly spending limits that match the customer's preferences.

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# customer wants limit >= 50,000
filtered_limit_cards = bank_df.filter(col
("Monthly_Limit") >= 50000) \
    .select
    ("Bank_Name",
    "Card_Name",
    "Monthly_Limit")

filtered_limit_cards.display()

(1) Spark Jobs

filtered limit cards: pyspark.sql.dataframe.DataFrame = [Bank Name: string, Card Name: string ... 1 more field]

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	Bank_Name	Card_Name	Monthly_Limit
1	ICICI	platinum ship credit card	60000
2	ICICI	Amazon pay icici card	100000
3	ICICI	Coral Credit Card	80000
4	IDFC	Millennia	125000
5	IDFC	First Classic	150000
6	IDFC	WOW ! Black	80000
7	IDFC	First Select	150000
8	IDFC	Ace	75000
9	IDFC	Republic Day	50000
10	PNB	Global Platinum	1000000
11	PNB	Global Gold	80000
12	PNB	Global Classic	80000
13	PNB	Millennial	400000
14	PNB	Patanjali RuPay	80000

8. Find Cards with Highest Rewards Points

- Goal: Find credit cards that provide the highest reward points for every dollar spent.
- Input: Customer's monthly expenditure across different categories.
- Output: List of credit cards offering the highest reward points.

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10:59 AM (<1s) 20
reward_cards = bank_df.orderBy(desc("Rewards
Points")) \
    .select("Bank_Name",
            "Card_Name", "Rewards
            Points")

reward_cards.display(5)
(1) Spark Jobs

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	Bank_Name	Card_Name	.00 Rewards Points		
1	Axis Bank	Axis Bank Neo Credit Card	200.0000		
2	IDFC	Millennia	150.0000		
3	PNB	Patanjali RuPay	150.0000		
4	PNB	Global Platinum	150.0000		
5	PNB	Global Gold	100.0000		
6	PNB	Millennial	100.0000		
7	PNB	Global Classic	50.0000		
8	HDFC Bank	Infinia Metal	10.0000		
9	HDFC Bank	Freedom	10.0000		
10	HDFC Bank	Diners Club Black Metal	10.0000		
11	Kotak Mahindra Bank	Zen Signature Credit Card	6.0000		
12	Canara Bank	RuPay Platinum Credit Card	5.0000		
13	Kotak Mahindra Bank	League Platinum Credit Card	5.0000		

9. Lounge Access Preferences

- Goal: Identify credit cards that offer lounge access and match the customer's preferences for travel benefits.
- Input: Customer's travel habits and lounge access requirements.
- Output: List of credit cards with lounge access, categorized by frequency per year.

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11:00 AM (<1s) 22 Python
lounge_cards = bank_df.filter(col("Lounge access
per year") > 0) \
    .select("Bank_Name",
            "Card_Name", "Lounge
            access per year") \
    .orderBy(desc("Lounge
            access per year"))

lounge_cards.display()
▶ (1) Spark Jobs

```

	^A _C Bank_Name	^A _C Card_Name	¹ ₃ Lounge access per year
1	SBI	SBI Card ELITE	14
2	SBI	SBI Card Prime	12
3	HDFC Bank	Classic Platinum	8
4	PNB	Global Platinum	8
5	HDFC Bank	Preferred Platinum	8
6	HDFC Bank	Platinum Debit	8
7	HDFC Bank	Vishesh Debit	8
8	Federal Bank	Mastercard Imperio	6
9	IDFC	First Classic	6
10	IDFC	First Select	6
11	Federal Bank	Visa Imperio	6
12	Canara Bank	Canara Gold Credit Card	5
13	Canara Bank	RuPay Platinum Credit Card	5
14	IDFC	Millennia	5
15	SBI	SBI Platinum International Debit Ca...	4

10. Customer-Specific Credit Card Recommendations

● Goal: Provide personalized credit card recommendations based on a customer's profile,

which includes income, spending habits (grocery, dining, travel), and reward preferences.

● Input: Customer's annual income, monthly spending in different categories (grocery, travel, dining), and rewards/benefits preference.

● Output: Recommended credit card(s) that align with the customer's needs.

Approach

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from pyspark.sql.functions import desc

recommendation_df = cashback_df.withColumn(
    "Score",
    col("Total_Cashback") +
    col("Rewards Points") * 0.1 +
    col("Lounge access per year") * 50 -
    col("Annual_Fee")
)

final_recommendations = recommendation_df.select(
    "Name", "Card_Name", "Score"
).orderBy("Name", desc("Score"))

final_recommendations.display()
▶ (2) Spark Jobs

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	A ^B _C Name	A ^B _C Card_Name	1.2 Score
1	Aarav Agarwal	Axis Bank Delight Debit Card	3370
2	Aarav Agarwal	Coral Credit Card	3307.9519999999...
3	Aarav Agarwal	Prime Credit Card	3160.1
4	Aarav Agarwal	HPCL Credit Card	2861.1
5	Aarav Agarwal	IRCTC Credit Card	2860.1
6	Aarav Agarwal	Prime Credit Card	2769.35
7	Aarav Agarwal	Prime Credit Card	2760.75
8	Aarav Agarwal	IDBI Delight Credit Card	2574.0499999999...
9	Aarav Agarwal	HPCL Credit Card	2470.35
10	Aarav Agarwal	IRCTC Credit Card	2469.35
11	Aarav Agarwal	HPCL Credit Card	2461.75
12	Aarav Agarwal	IRCTC Credit Card	2460.75
13	Aarav Agarwal	Coral Credit Card	2445.302
14	Aarav Agarwal	Axis Bank Delight Debit Card	2268.75